

WTW158 Calculus

Class Test 1 (Monday)

20 - 24 May 2024

Time: 15 min

8

Surname: Initials: Student number: Group: M-number:

Use a pen to complete the test. Show all your steps and simplify your answers. Use interval notation, where applicable. You are not allowed to use a calculator. The test is printed on both pages.

QUESTION 1 [1 mark]

Suppose f'' is continuous near c . Complete the following statement of the Second Derivative Test:

If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at $x=c$.

QUESTION 2 [1]

Write down (only) the form of the limit. Use the space provided.

(i) $\lim_{x \rightarrow 0^+} (2x)^{\frac{1}{\ln x}}$ (0^0)

(ii) $\lim_{x \rightarrow 0^+} (1 + \sin 3x)^{\frac{1}{x}}$ (1^∞)

QUESTION 3 [2 marks]

Determine the limit (if it exists):

$$\lim_{x \rightarrow 0^+} x \ln x$$

$$\lim_{x \rightarrow 0^+} x \ln x \quad (0 \times (-\infty))$$

$$= \lim_{x \rightarrow 0^+} \frac{\ln x}{\frac{1}{x}} \quad \left(\frac{-\infty}{\infty} \right)$$

$$= \lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{\frac{-1}{x^2}} \quad (\text{L'Hospital})$$

$$= \lim_{x \rightarrow 0^+} \frac{1}{x} \left(\frac{x^2}{-1} \right)$$

$$= \lim_{x \rightarrow 0^+} x = 0.$$

QUESTION 4 [2.5, 1.5]

Given:

$$f(x) = \frac{1}{5}x^5 - \frac{8}{3}x^3 + 16x$$

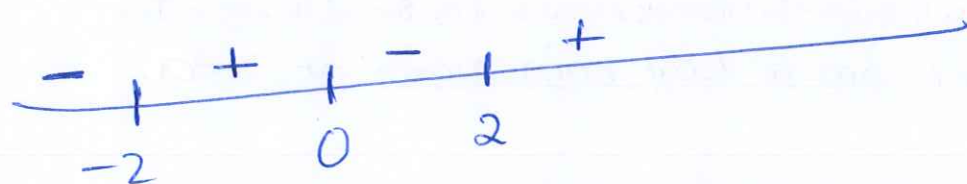
$$f'(x) = (x^2 - 4)^2$$

$$f''(x) = 4x(x^2 - 4).$$

- (i) Show how you use the Concavity Test to determine the interval(s) on which f is concave upward and concave downward.

Sign of f''(x) to determine concavity

Find the sign of $f''(x)$



$$\begin{aligned} f''(-10) &< 0 \\ f''(-1) &> 0 \\ f''(1) &< 0 \\ f''(10) &> 0 \end{aligned}$$

$f''(x) < 0$ on $(-\infty, -2)$ and on $(0, 2)$

$\Rightarrow f$ is concave downward on $(-\infty, -2) \cup (0, 2)$

$f''(x) > 0$ on $(-2, 0)$ and on $(2, \infty)$

$\Rightarrow f$ is concave upward on $(-2, 0)$ and $(2, \infty)$.

- (ii) Show how you use the Increasing/Decreasing Test to determine the interval(s) on which f is increasing.

f is increasing where $f'(x) > 0$

$$f'(x) = (x^2 - 4)^2$$

$$f'(x) > 0 \text{ on all } x \in \mathbb{R} \setminus \{-2, 2\}$$

$$\Rightarrow f'(x) > 0 \text{ on } (-\infty, -2) \cup (-2, 2) \cup (2, \infty)$$

Ans: f is increasing on $(-\infty, -2) \cup (-2, 2) \cup (2, \infty)$.

MEMO

WTW158 Calculus

Class Test 5 (Tuesday)

20 - 24 May 2024

Time: 15 min

8

Surname:

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Initials:

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Student number:

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Use a pen to complete the test. Show all your steps and simplify your answers. Use interval notation, where applicable. You are not allowed to use a calculator. The test is printed on both pages.

QUESTION 1 [1]

Complete the following part of the Concavity Test:

If $f''(x) > 0$ on the interval I , then the graph of f is concave upward on I .

QUESTION 2 [1.5, 2.5]

Given: $f(x) = \frac{x}{x^2 - 4}$, with $f'(x) = \frac{-(x^2 + 4)}{(x^2 - 4)^2}$, and $f''(x) = \frac{2x(x^2 + 12)}{(x^2 - 4)^3}$

(i) Determine the interval(s) on which $f(x)$ decreasing.

f is decreasing where $f'(x) < 0$

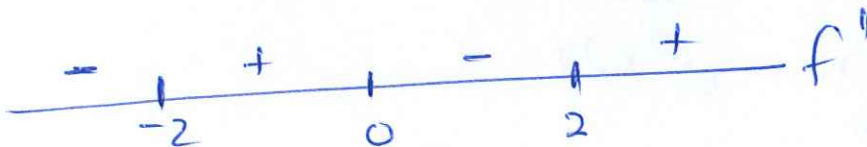
$f'(x) = \frac{-(x^2 + 4)}{(x^2 - 4)^2}$ is negative on its domain

$\Rightarrow f$ is decreasing on $(-\infty, -2) \cup (-2, 2) \cup (2, \infty)$.

(ii) Determine the interval(s) on which $f(x)$ is concave upward and concave downward.

$$f''(x) = \frac{2x(x^2 + 12)}{(x^2 - 4)^3}$$

Sign of f'' :



$$f''(-10) < 0$$

$$f''(-1) > 0$$

$$f''(1) < 0$$

$$f''(10) > 0$$

$f''(x) > 0$ on $(-2, 0)$ and $(2, \infty)$

$\Rightarrow f$ is concave upward on $(-2, 0)$ and $(2, \infty)$

$f''(x) < 0$ on $(-\infty, -2)$ and $(0, 2)$

$\Rightarrow f$ is concave downward on $(-\infty, -2)$ and $(0, 2)$.

QUESTION 3 [1]

Give (only) the form of the limits:

(i) $\lim_{x \rightarrow 0^+} (\cos x)^{\frac{1}{x^2}}$

 (1^∞)

(ii) $\lim_{x \rightarrow -\infty} x^4 + x^3$

 $(\infty - \infty)$ **QUESTION 4 [2]**

Determine the limit:

$$\lim_{x \rightarrow \infty} x^3 e^{-x^2}$$

$$\lim_{x \rightarrow \infty} x^3 e^{-x^2} \quad (\infty \times 0)$$

$$= \lim_{x \rightarrow \infty} \frac{x^3}{e^{x^2}} \quad \left(\frac{\infty}{\infty}\right)$$

$$\stackrel{\text{L'Hospital}}{=} \lim_{x \rightarrow \infty} \frac{3x^2}{e^{x^2} \times 2x}$$

$$= \frac{3}{2} \lim_{x \rightarrow \infty} \frac{x}{e^{x^2}} \quad \left(\frac{\infty}{\infty}\right)$$

$$\stackrel{\text{L'Hospital}}{=} \frac{3}{2} \lim_{x \rightarrow \infty} \frac{1}{e^{x^2} \times 2x}$$

$$= 0$$

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Class Test 5 (Thursday)

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Use a pen to complete the test. Show all your steps and simplify your answers. Use interval notation, where applicable. You are not allowed to use a calculator. The test is printed on both pages.

QUESTION 1 [1]

Complete the sentence: Suppose that c is a critical number of a continuous function f . If f' changes from positive to negative at c , then f has a local maximum at $x=c$.

QUESTION 2 [2]

Given the functions below. Use the second derivative test to determine the local maximum/minimum (if any) of f .

$$f(x) = x^2 - x - \ln x, f'(x) = \frac{(2x+1)(x-1)}{x} \text{ and } f''(x) = \frac{1+2x^2}{x^2}$$

$$f'(x) = 0 \Rightarrow \frac{(2x+1)(x-1)}{x} = 0 \Rightarrow (2x+1)(x-1) = 0$$

$$\Rightarrow x = -\frac{1}{2} \text{ or } x = 1$$

But $x > 0 \Rightarrow x = 1$ is the only critical number.

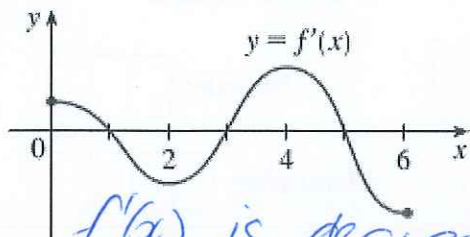
$$f''(1) = \frac{1+2}{1} = 3 > 0$$

$\Rightarrow f(1)$ is the local minimum of f

$f(x)$ has ~~not~~ no local maximum.

QUESTION 3 [2 marks]

Given the graph of $f'(x)$ defined on $(0, 6)$ below. Use the concavity test to determine the open interval(s) on which $f(x)$ is concave downward. Motivate your answer.



$f(x)$ is concave downward
where $f''(x) < 0$

$f'(x)$ is decreasing on $(0, 2)$ and on $(4, 6)$

$\Rightarrow f''(x) < 0$ on $(0, 2)$ and on $(4, 6)$

$\Rightarrow f$ is concave downward on $(0, 2)$ and $(4, 6)$.

QUESTION 4 [3 marks]

Evaluate the limit and also state the form where appropriate.

$$\lim_{x \rightarrow \infty} (1 + \frac{2}{x})^x$$

$$\lim_{x \rightarrow \infty} (1 + \frac{2}{x})^x \quad (1^\infty)$$

$$\text{Let } y = (1 + \frac{2}{x})^x$$

$$\text{Then } \ln y = \ln(1 + \frac{2}{x})^x = x \ln(1 + \frac{2}{x})$$

$$\lim_{x \rightarrow \infty} \ln y = \lim_{x \rightarrow \infty} x \ln(1 + \frac{2}{x}) \quad (\infty \times 0)$$

$$= \lim_{x \rightarrow \infty} \frac{\ln(1 + \frac{2}{x})}{\frac{1}{x}} \quad (\frac{0}{0})$$

$$\begin{aligned} &\text{L'Hosp.} \\ &= \lim_{x \rightarrow \infty} \frac{(\frac{1}{1 + \frac{2}{x}})(-\frac{2}{x^2})}{(-x^{-2})} \end{aligned}$$

$$= 2 \lim_{x \rightarrow \infty} \frac{1}{1 + \frac{2}{x}} = 2$$

$$\Rightarrow \lim_{x \rightarrow \infty} y = \lim_{x \rightarrow \infty} e^{\ln y} = e^{\lim_{x \rightarrow \infty} \ln y} = e^2.$$

WTW158 Calculus

Class Test 5 (Friday)

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Use a pen to complete the test. Show all your steps and simplify your answers. Use interval notation, where applicable. You are not allowed to use a calculator. The test is printed on both pages.

QUESTION 1 [1]

Complete the definition: If the graph of f lies above all of its tangents on an interval I , then f is called

concave upward on I .

QUESTION 2 [3]

Evaluate the limit and also state the form where appropriate.

$$\lim_{x \rightarrow 0^+} x^{\sqrt{x}} \quad (0^0)$$

$$\text{Let } y = x^{\sqrt{x}}$$

$$\text{Then } \ln y = \ln x^{\sqrt{x}} = \sqrt{x} \ln x$$

$$\lim_{x \rightarrow 0^+} \ln y = \lim_{x \rightarrow 0} \sqrt{x} \ln x \quad (0 \times -\infty)$$

$$= \lim_{x \rightarrow 0} \frac{\ln x}{x^{-1/2}} \quad \left(\frac{-\infty}{\infty} \right)$$

$$\text{L'Hosp.} = \lim_{x \rightarrow 0} \frac{\frac{1}{x}}{-\frac{1}{2} x^{-3/2}}$$

$$= -2 \lim_{x \rightarrow 0} \frac{1}{x} \times x^{3/2} = -2 \lim_{x \rightarrow 0} x^{1/2} = -2 \times 0 = 0$$

$$\Rightarrow \lim_{x \rightarrow 0^+} x^{\sqrt{x}} = \lim_{x \rightarrow 0^+} e^{\ln y} = e^{\lim_{x \rightarrow 0^+} \ln y} = e^0 = \underline{\underline{1}}$$

QUESTION 3 [2]

Suppose

$$f(x) = x^4 e^{-x}, f'(x) = x^3 e^{-x}(4-x) \text{ and } f''(x) = e^{-x}x^2(12+x^2).$$

Use the first derivative test to determine the local minimum of f .

$$f'(x) = 0 \Rightarrow x^3 e^{-x}(4-x) = 0 \Rightarrow x=0 \text{ or } x=4.$$

Critical numbers: $x=0$ and $x=4$.



$$f''(-10) < 0$$

$$f'(1) > 0$$

$$f'(10) < 0$$

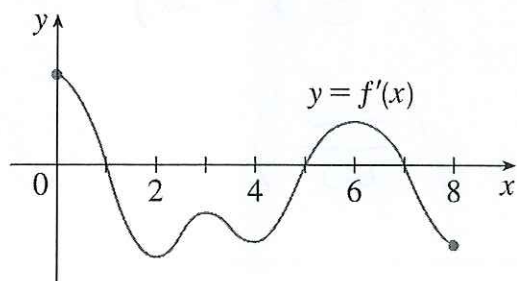
$\Rightarrow f'$ changes from negative to positive at $x=0$

$\Rightarrow f$ has a local minimum at 0, $f(0) = 0$.

QUESTION 4 [2]

Use the graph of $f'(x)$ defined on $[0, 8]$ to answer the question.

At which x -values does f have a local maximum. Use the second derivative test and give reasons for your answer.



f has a local maximum where $f'(x) = 0$

$$f'(x) = 0 \text{ at } x=1, x=5 \text{ and } x=7.$$

$f'(x)$ decreases on $(0, 2)$ and on $(6, 8)$
 $\Rightarrow f''(x) < 0$ on $(0, 2)$ and on $(6, 8)$.

$$f''(1) < 0 \text{ and } f''(7) < 0$$

$\Rightarrow f$ has a local maximum at $x=1$ and $x=7$.